

Dividend Factors

Firm Specific Factors

- Firm's internal constraints and restrictions
- Magnitude and earnings trend
- Future financial requirements
- Type of shareholders
- Nature of industry
- Age of the firm
- Shareholder's considerations
- Requirements of institutional investors
- Control objectives

External Factors

- Legal contractual
- Capital market considerations
- Government's economic policy
- Inflation
- Taxation policy
- Industry Dividend pay-out ratio
- Stability of dividends in the economy

Dividend Policy

- Residual Dividend Approach
- Dividend Stability
- Compromise Dividend Policy

Residual Dividend Approach

- A policy under which a firm pays dividends only after meeting its investment needs while maintaining a desired debt-equity ratio.
- Firm expects to minimize the need to sell new equity, hence does not sell new equity to pay out dividend (expensive).
- Firm relies on internally generated equity to finance new positive NPV projects.
- Firm's objective is to meet its investment needs and maintain its desired debt-equity ratio before paying dividends.

Example

- Firm has 1,000 in earnings and debt equity ratio of 1:2. Assume the firm needs 900 for new projects.
 - If the firm reinvests the entire 1,000 and pays no dividend, then equity will increase by 1,000.
 - To main debt equity ratio, firm borrows additional amount of 500
 - Total amount of funds that can be generated without selling new equity is thus $1,000 + 500 = 1,500$
 - Firm must generate 900 (1,000 earnings, but the firm to maintain the current capital structure has to borrow)
 - Firm will borrow $1/3 \times 900 = 300$
 - Remaining 600 from earnings of 1,000
 - Residual fund available for dividend payment = $1,000 - 600 = 400$

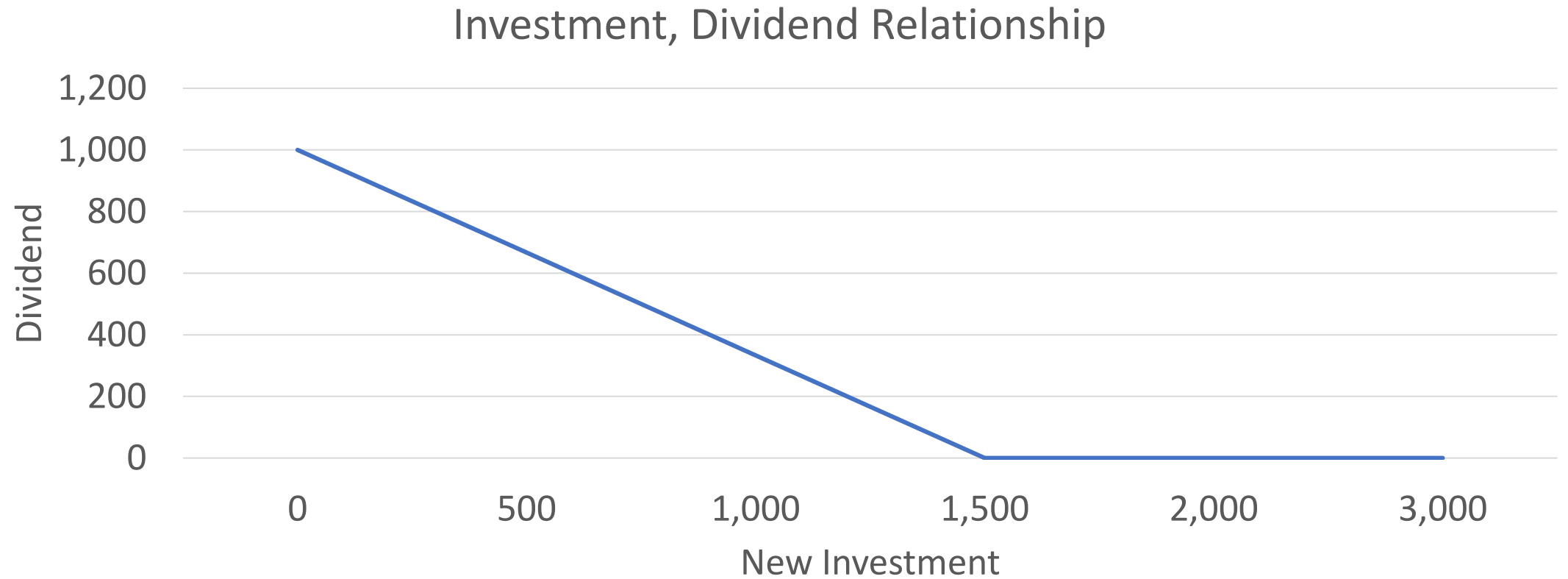
Exercise

S.No	Aftertax Earnings	New Investment	Additional Debt	Retained Earnings	Additional Share Value	Dividend
1	1,000	3,000	?	?	?	?
2	1,000	2,000	?	?	?	?
3	1,000	1,500	?	?	?	?
4	1,000	1,000	?	?	?	?
5	1,000	500	?	?	?	?
6	1,000	0	?	?	?	?

Exercise

S.No	Aftertax Earnings	New Investment	Additional Debt	Retained Earnings	Additional Share Value	Dividend
1	1,000	3,000	1,000	1,000	1,000	0
2	1,000	2,000	667	1,000	333	0
3	1,000	1,500	500	1,000	0	0
4	1,000	1,000	333	667	0	333
5	1,000	500	167	333	0	667
6	1,000	0	0	0	0	1,000

Investment / Dividend Relationship



Dividend Payment Policy Criteria

- A firm with many investment opportunities will pay small amounts of dividends and a firm with few investment opportunities will pay relatively larger amounts of dividends
- Younger, fast-growing firms have a low payout ratio, whereas older, slower-growing firms in more mature industries use a higher payout ratio.
- Higher the dividend payout ratio, lower the proportion of earnings requirements towards new investment, hence higher the capacity to pay dividends.

Residual Dividend Policy Equation

- **Dividend or Equity Issue = Net Income – [Capital Budget x Target equity ratio]**
- *Where, target equity ratio is the proportion of equity in the capital structure.*
- Net income higher than proportional capital budget leads to dividend payment
- Net income lower than proportional capital budget leads to new equity issue (external equity financing).

Example

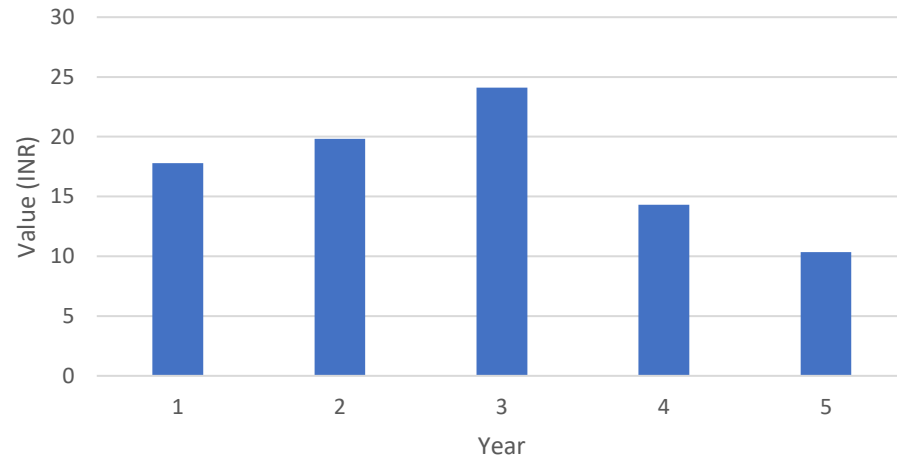
- Company X has a residual decision policy. It expects to generate INR 2 million in net earnings after taxes in the coming year. The company has an all-equity capital structure and cost of equity capital is 15%. Including floatation and underwriting costs, the equity financing cost is higher at 16%.
- Firm has INR 1.5 million project proposals.
- Dividend payment = $(2 - 1.5) = \text{INR } 0.5 \text{ million}$
- Firm has INR 2 million project proposals.
- Dividend payment = $(2 - 2) = \text{INR } 0 \text{ million}$
- Firm has INR 3 million project proposals.
- Dividend payment = $(2 - 3) = \text{INR } 0 \text{ million}$ Equity issue required = INR 1 million

Dividend Stability

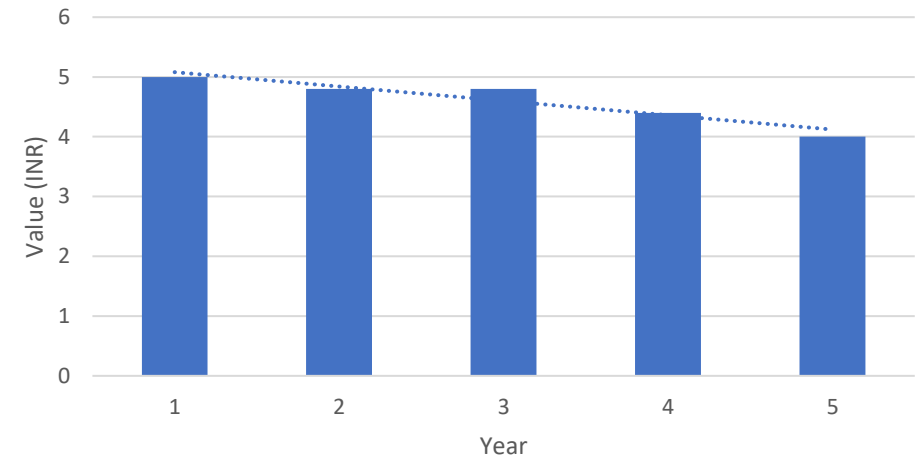
- Strict residual dividend policy tends to project an unstable dividend policy.
 - If investment opportunities in one period are quite high, dividends will be low or zero.
 - Conversely, dividends might be high in the next period of investment opportunities are considered less promising.
- **Stable Dividend Policy**
 - Each quarter's dividend is a fixed fraction of yearly earnings
 - All dividend payments will be equal
- Earnings also exhibit a seasonal trend
 - Quarterly earnings vary throughout the year.
- **Cyclical Dividend Policy**
 - Dividends are constant proportion of earnings at each pay date

Stable Dividend Policy

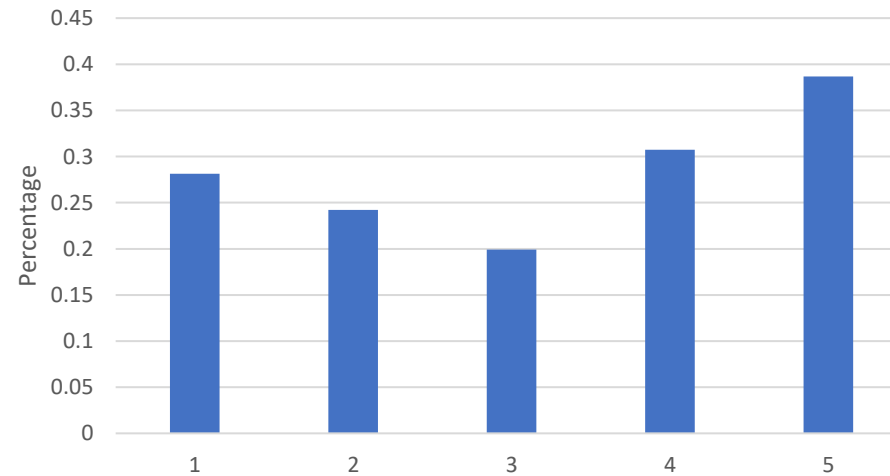
Earnings Per Share



Dividend Per Share

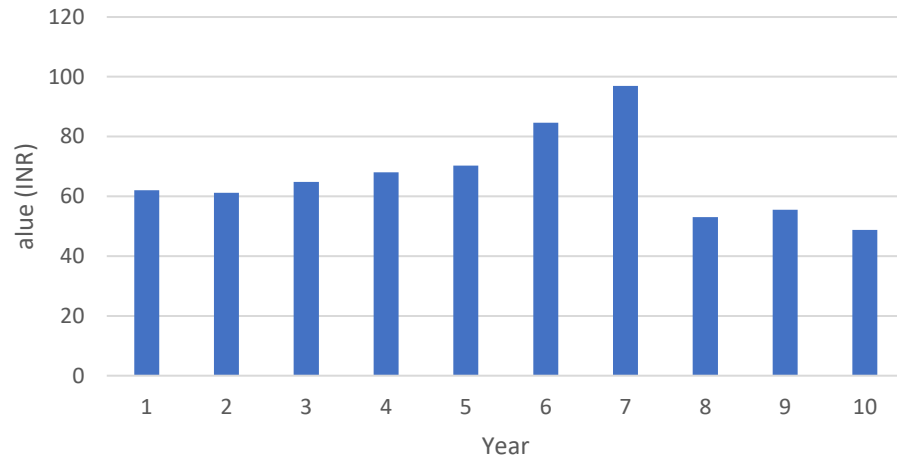


Dividend Payment Percentage

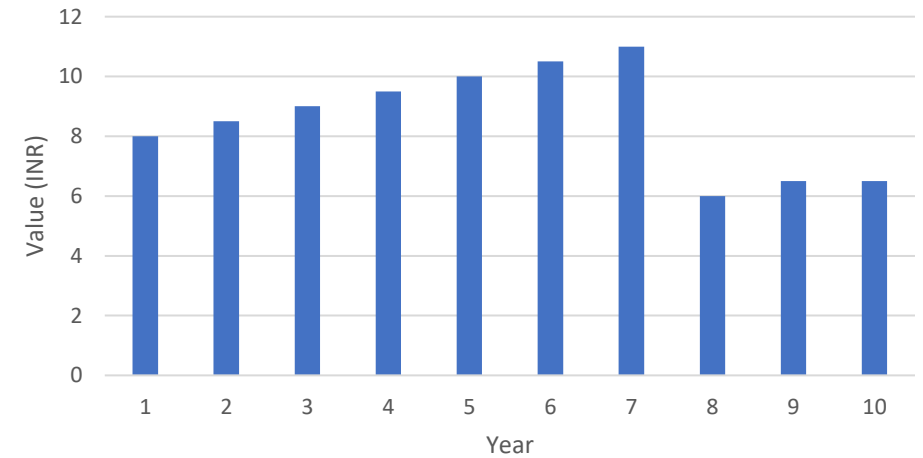


Stable Dividend Policy

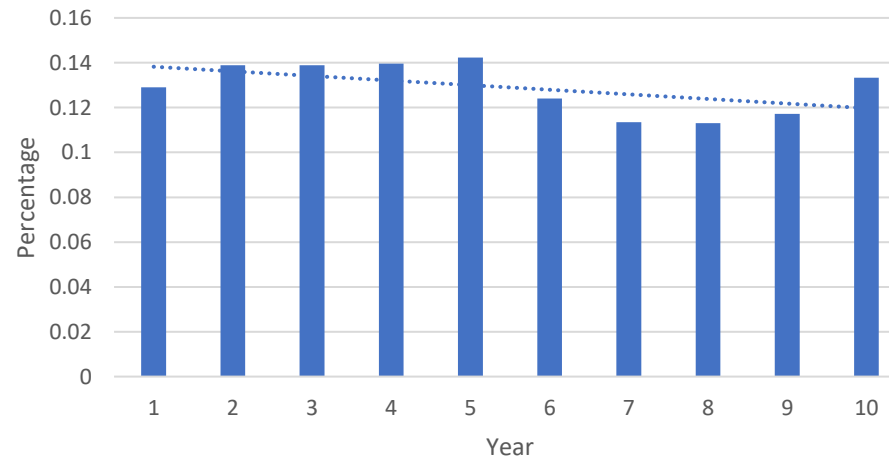
Earnings Per Share



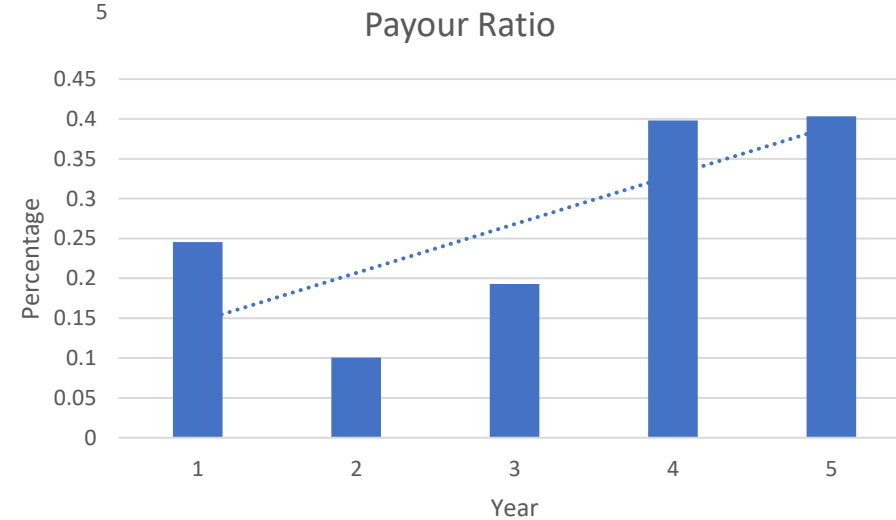
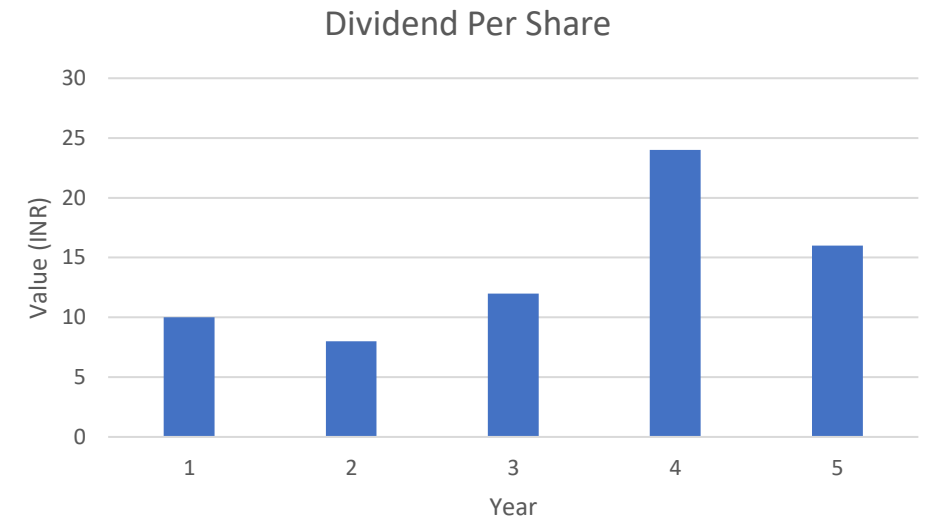
Dividend Per Share



Payout Ratio



Unstable Dividend Policy



Sticky Dividend

- Firms generally do not change their dividends frequently.
- Reluctance to change dividends results in sticky dividends
- Firm's concern about its capability to maintain higher dividends in future periods.
- Markets tend to take a negative view of dividend decreases, and the stock price drops to reflect the signals.
- Firms set target payout ratios
- Dividends lag earnings
- Dividend changes are infrequent

Compromise Dividend Policy

- Avoid cutting back on positive NPV projects to pay a dividend
- Avoid dividend cuts
- Avoid the need to sell equity
- Maintain a target debt-equity ratio
- Maintain a target dividend payout ratio.
 - Debt-Equity ratio is viewed as long-range goal, it is allowed to vary in short run if necessary to avoid a dividend cut or the need to sell the new equity.
- Minimize the problems of dividend instability by creating two types of dividends :
 - Regular Dividend
 - Extraordinary Dividend

Compromise Policy Considerations

- Given the volatility in earnings and cash flows at firms, dividends do not reflect that volatility
- Firms do not actively reassess how much they should pay in dividends.
- Firms set a level of dividends (payout ratios) by considering industry norms
- Focus on changes in dividends in response to changes in earnings.
- Apply simple rules of thumb on how to adjust dividends, such as raising dividends only if earnings increase 30% or more.
- Avoid adjusting dividends in response to changes in stockholder attitudes, if these changes are viewed as short-term changes.

Measures of Dividend Policy

- Dividend yield
 - Annual dividend per share / Price per share
 - Provides a measure of that component of the total return that comes from dividends, with the balance coming from price appreciation
 - Dividend yield is a measure of expected return on stock
 - Market price = Dividend yield + Price appreciation
 - Some investors also use the dividend yield as a measure of risk and as an investment screen, that is, they invest in stocks with high dividend yields.

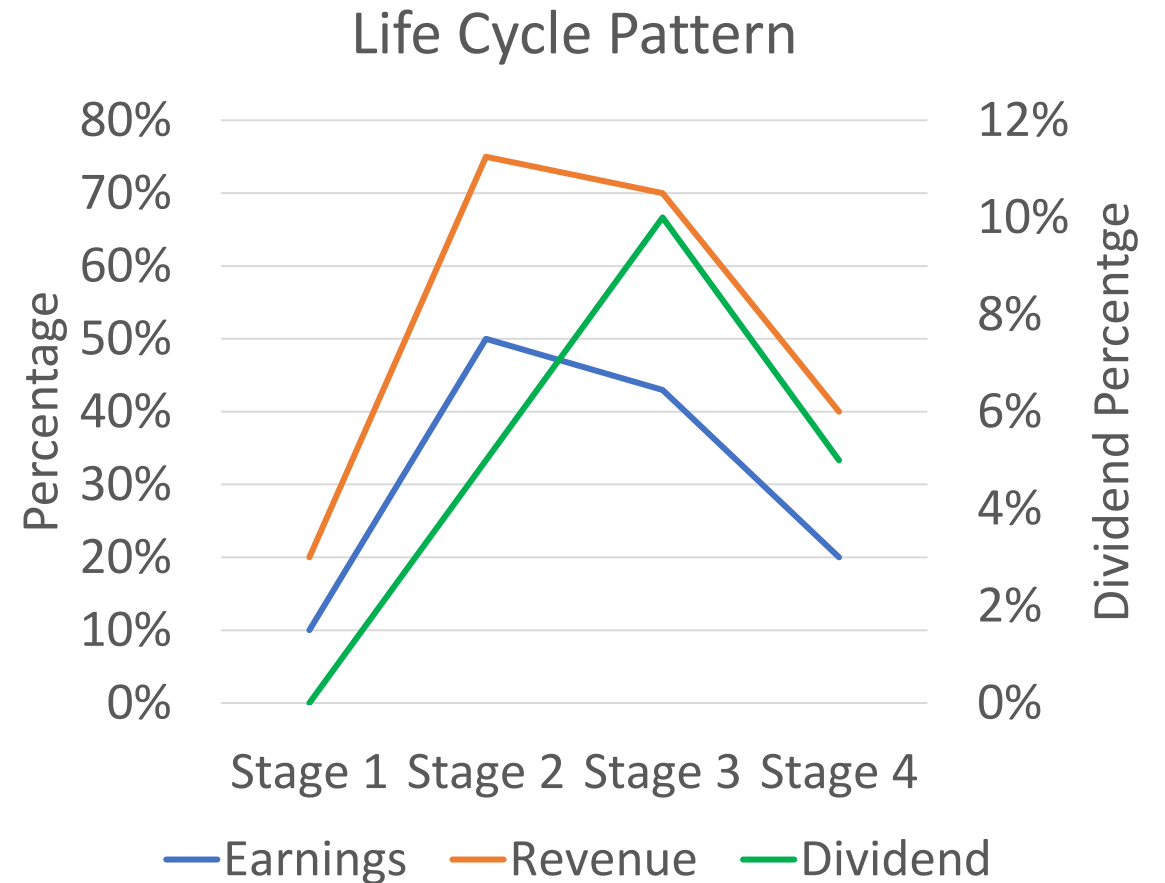
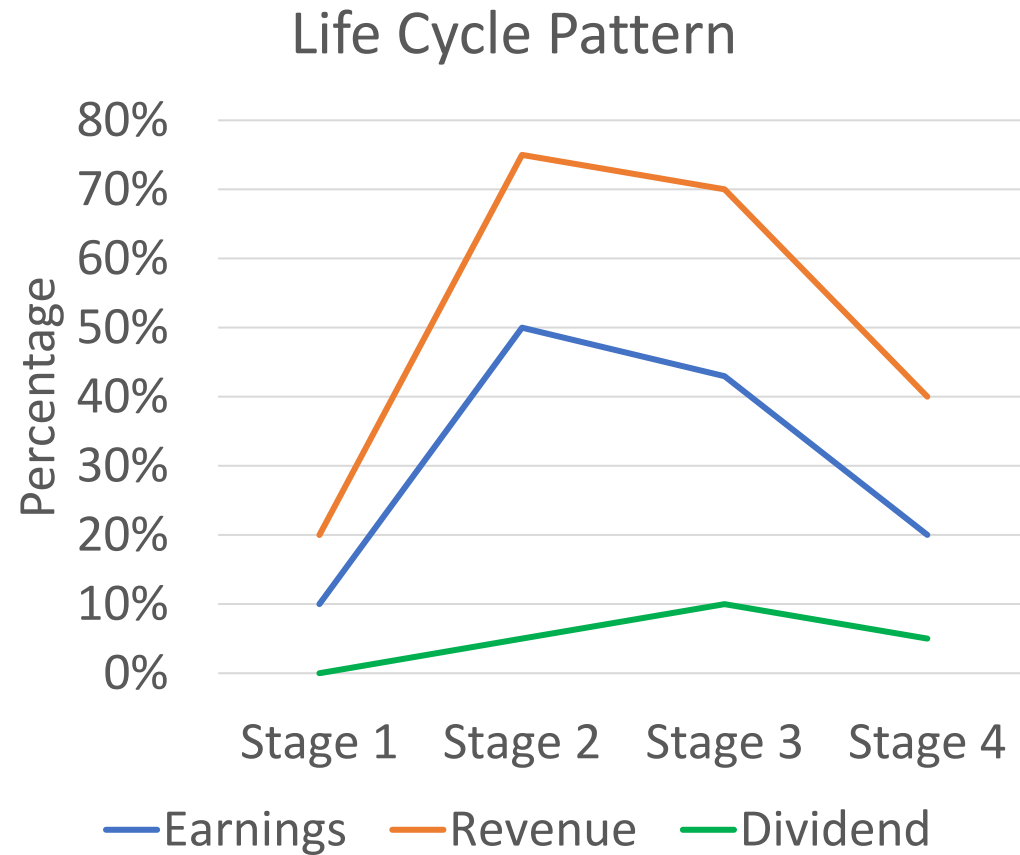
Measures of Dividend Policy

- Dividend payout ratio
 - Annual dividend paid / Annual earnings
 - Payout ratio is used in valuation as a way of estimating dividends in future periods (analysts estimate growth in earnings rather than dividends).
 - Retention ratio is the proportion of the earnings reinvested in the firm (Retention Ratio = $1 - \text{Dividend Payout Ratio}$): useful in estimating future growth in earnings (firms with high retention ratios (low payout ratios) have higher growth rates in earnings than firms with lower retention ratios (higher payout ratios))
 - Dividend payout ratio tends to follow the life cycle of the firm, starting at zero when the firm is in high growth and gradually increasing as the firm matures and its growth prospects decrease

Dividend Policy and Firm Life Cycle

	Stage 1 (Start-up)	Stage 2 (Expansion)	Stage 3 (Mature)	Stage 4 (Decline)
Fund Requirement	Limited	High	Moderate	Low
Retained Earnings	Negative	Low	Increases	High
Cashflow for dividends	Nil / very low	Nil / very low	Moderate	High
Growth Stage	Moderate	Very High	High	Low
Dividend Policy	Nil / Residual	Nil / Residual	Compromise	Stable

Firm's Life Cycle and Dividend



Ex- Vs Cum Dividend Equality

- Assume t_o represents the tax rate on dividends and t_{cg} represents the tax rate on capital gains. Let P_B denote the cum-dividend stock price, and P_A the ex-dividend stock price, and P the price at which the stock was acquired.
- For an investor, there is market choice of selling the shares at the ex or cum dividend price.

$$P_B - (P_B - P)t_{cg} = P_A - (P_A - P)t_{cg} + D(1 - t_o)$$

where:

- LHS is the after-tax gain from selling the stock cum-dividend
- RHS is the after-tax gain from selling the stock ex-dividend
- Equality =

$$\frac{P_B - P_A}{D} = \frac{1 - t_o}{1 - t_{cg}}$$

Tax Rate and Dividend Preference

- Empirical price drop (Ex-dividend less than Cum-dividend), implies that the marginal tax for holders of the firm's stock increases as the difference narrows.
- A necessary condition for the investor to be indifferent between selling before and after the ex-dividend day is that the price drop on the ex-dividend day must reflect the investor's tax differential between dividends and capital gains.

Income tax	Capital gains tax	Equality
0.3	0.15	82%
0.3	0.2	88%
0.3	0.25	93%

Dividend Preference across Scenarios

Market Scenario	Preference
$P_B - P_A = D$	Investor is indifferent between dividends and capital gains
$P_B - P_A < D$	Investor is taxed more heavily on dividends
$P_B - P_A > D$	Investor is taxed more heavily on capital gains

Corporate Tax Relief

- In some countries, firms are allowed to claim a partial deduction for dividends paid. This brings their dividend treatment into parity with the treatment of the interest paid on debt, which is entitled to a full deduction in most countries.
- Among the Organization for Economic Cooperation and Development (OECD) countries, the Czech Republic and Iceland offer partial deductions for dividend payments by firms.
- Germany until recently applied a higher tax rate to income that was retained by firms than to income that was paid out in dividends. In effect, this gives a partial tax deduction to dividends. (Partial exemption for dividend income from German corporations is 5%. Only 95% of dividend income is considered taxable.)
- Presence of foreign investors in the stock deters economies to stop the tax relief to the foreign shareholders (Withholding tax).
- Investors in the stock may be tax-exempt or pay no taxes, which effectively reduces the overall taxes paid on dividends to the treasury to zero.

Individual Tax Relief

- Tax credit for taxes paid by firms (taxed at source):
 - Individuals can be allowed to claim the taxes paid by the firms as a credit when computing their own taxes.
 - Example: Firm paid 30 percent of its income of INR 100 million as taxes and then paid its entire income as dividends to individuals with 40 percent tax rates. The individuals would be allowed to claim a tax credit of INR 30 million against the taxes owed, thus reducing taxes paid to INR 10 million. In effect, this will mean that only individuals with marginal tax rates that exceed the corporate tax rate will be taxed on dividends.
 - Australia, Finland, Mexico, and New Zealand allow individuals to get a full credit for corporate taxes paid. Canada, France, the United Kingdom, and Turkey allow for partial tax credits.
- Lower tax rate on dividends:
 - Dividends get taxed at a lower rate than other income to reflect the fact that it is paid out of after-tax income.
 - In some countries, the tax rate on dividends is set equal to the capital gains tax rate.
 - South Korea, for instance, has a flat tax rate of 16.5 percent for dividend income. This is the path that the United States chose in 2003 to grant relief from double taxation to stock investors.

Dividend Payment Preference

- Firms with an investor base composed primarily of individuals typically should have paid lower dividends than do firms with investor bases predominantly made up of tax-exempt institutions.
- Higher the income level (and hence the tax rates) of the investors holding stock in a firm, lower the dividend paid out by the firm.
- As the tax disadvantage associated with dividends increased, the aggregate amount paid in dividends should have decreased.
- Conversely, if the tax disadvantage associated with dividends decreased, the aggregate amount paid in dividends should have increased.

Tax Implications of Dividend

- Individual investors are subject to income tax.
 - Dividends high higher tax rate than capital gains because of the tax-timing option (ability for investors to postpone the tax liability on capital gains income)
 - Individual investors may prefer capital gains than dividend
- Corporate shareholders pay income tax at a higher marginal rate but are permitted to claim a dividends received deduction.
 - Marginal tax rate on dividend income for a corporation hence is lower (deduction on dividend received)
 - Corporate investors prefer dividends.
- Tax-exempt institutions, such as pension funds, do not have a bias in favor of capital gains or dividends

	Preference
High Tax Bracket Individual Investors	Zero or low dividend payment
Low Tax Bracket Individual Investors	High dividend payment
Tax-free Institutional Investors	Medium dividend payment
Corporate Investors	High dividend payment

Investor Psychology and Dividend Preference

- Absence of self control
 - To the extent that investors have trouble controlling consumption and resisting temptation, they look for simple rules that prevent them from indulgence. One simple rule with stocks that protects investors from overspending may be to consume the dividend but leave the principal untouched. Hence, dividend preference.
- Mental Accounting
 - With some utility functions, the utility gained by investors from a gain that is broken down into dividends and capital gains may be greater than the utility from the same gain, if delivered entirely as a capital gain. For instance, investors may get more utility when they receive INR 10 in dividends and INR 40 in capital gains than from a capital gain of INR 50.
- Regret avoidance
 - Investors regret mistakes, but they regret errors of commission more than errors of omission. An investor who buys a non-dividend stock that goes down may be forced to sell the stock to generate cash and is thus forced to confront his or her error. In contrast, an investor who buys a dividend paying stock that goes down may be able to get by without selling the stock and feels less regret

Dividend as Information Signal

- Financial markets view announcements made by firms about their future prospects since firms routinely make exaggerated claims.
- Firms with good investment prospects are undervalued by markets.
- Signaling theory suggests that firms need to take actions that cannot be easily imitated by firms without good projects.
 - By increasing dividends, firms create a cost to themselves, because they commit to paying these dividends in the long run. Their willingness to make this commitment indicates to investors that they believe they have the capacity to generate these cash flows in the long run. This positive signal lead investors to reevaluate the cash flows and firm values and increase the stock price.
 - Decreasing dividends is a negative signal, largely because firms are reluctant to cut dividends. Thus, when a firm takes this action, markets see it as an indication that this firm is in substantial, long-term financial trouble. Consequently, such actions lead to a drop in stock prices.

Dividend Preference and Firm Leverage Decision

- When dividends increase, stockholders sometimes get a payment in the form of a wealth transfer from lenders to the firm.
- Payment of dividends takes cash out of the firm and this cash could have been used to cover outstanding interest or principal payments.
- Lenders would rather have firms accumulate cash than pay it out as dividends.
- Bond prices decline on the announcement of large increases in dividends. Bondholders protect themselves against this loss by restricting how much firms can pay out in dividends.
- Equity investors gain from the loss in market value faced by bondholders.

Managerial Preference for Dividend

- Managerial over confidence
 - Decisions made by managers can be affected by their confidence. In addition to borrowing too much money, issuing too little new equity, over-estimating the cash flows and benefits from acquisitions and investing in too many projects, over-confident managers also tend to pay too little in dividends. Dividend payout ratios at firms run by overconfident CEOs are lower than otherwise similar firms run by less confident CEOs.
- Conservative vs Aggressive managers
 - Management predispositions follow them from firm to firm. Conservative managers tend to pay less dividends and accumulate more cash than aggressive managers of firms.